Network Dynamics – Homework 1

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EXERCISE 1

Part (a)

A cut in the network is a partition of the nodes into two sets, where the origin o belongs to the first set and the destination d belongs to the second one. The capacity of a cut is defined as the sum of the capacities of all edges going from the first set to the second. By evaluating all possible o–d cuts of the given network, we can determine how much capacity must be removed in order to disconnect the origin from the destination.

We consider the following cuts:

- $U = \{o\}$, $U^c = \{a, b, c, d\}$: cut edges (o, a) and (o, b), with total capacity $3 + 3 = 6$.
- $U = \{o, a\}$, $U^c = \{b, c, d\}$: cut edges (o, b), (a, d) and (a, b), with total capacity $3 + 3 + 1 = 7$.
- $U = \{o, b\}$, $U^c = \{a, c, d\}$: cut edges (o, a), (b, c) and (a, b), with total capacity $3 + 3 + 1 = 7$.
- $U = \{o, a, b\}$, $U^c = \{c, d\}$: cut edges (b, c), (a, d) and (c, d), with total capacity $3 + 3 + 2 = 8$.
- $U = \{o, b, c\}$, $U^c = \{a, d\}$: cut edges (o, a), (c, d) and (a, b), with total capacity $3 + 2 + 1 = 6$.
- $U = \{o, a, b, c\}$, $U^c = \{d\}$: cut edges (a, d) and (c, d), with total capacity $3 + 2 = 5$.

Among all cuts, the smallest one is

$$U = \{o, a, b, c\}, \qquad C(U) = 5.$$

Therefore, the minimum capacity that must be removed to disconnect o from d is

$$C^*_{o,d} = 5,$$

which corresponds to removing the capacity across edges (a, d) and (c, d), forming the minimum o–d cut.

Part (b)

From Part (a) we know that the minimum o–d cut has capacity

$$C^*_{o,d} = 5,$$

and it is obtained by separating node d from the rest of the network. In this case the edges crossing the cut are (a, d) and (c, d) with capacities 3 and 2, respectively. Since the maximum throughput that can be sent from o to d is equal to the capacity of the minimum cut (Max–Flow Min–Cut theorem), the initial maximum throughput is $T(0) = 5$.

We are given $x > 0$ extra units of capacity and we want to distribute them on the links in order to maximize the throughput. The idea is to always allocate capacity on edges that belong to a minimum cut, because increasing the capacity of those edges is the only way to increase the value of the minimum cut and thus the maximum flow.

In our network, the edges (a, d) and (c, d) are the main bottleneck: they appear in the minimum cut and they are the only links directly connecting the intermediate nodes to the destination. Therefore, the best strategy is to progressively add the extra capacity to these two links. After each allocation we recompute the minimum cut. By doing this iteratively, we observe the following behaviour:

- the first extra unit of capacity increases the minimum cut from 5 to 6;
- the second extra unit does not increase the throughput further, because another cut becomes critical;
- from that point on, every two additional units of capacity lead to an increase of one unit in the maximum throughput.

As a consequence, the maximum throughput as a function of the extra capacity $x \in \mathbb{Z}_{\geq 0}$ can be written as

$$T(x) = 5 + \left\lfloor \frac{x+1}{2} \right\rfloor.$$

For $x = 0, 1, 2, \ldots, 10$ we obtain

$$T(x) = 5,\ 6,\ 6,\ 7,\ 7,\ 8,\ 8,\ 9,\ 9,\ 10,\ 10,$$

which corresponds to a staircase-shaped plot where the throughput increases by one unit approximately every two units of added capacity.

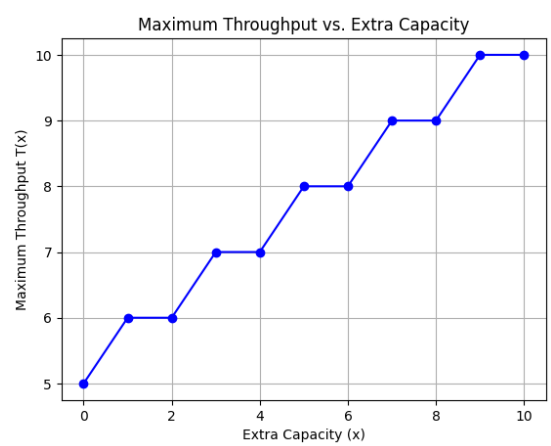

Fig. 1. Maximum Throughput vs. Extra Capacity

Part (c)

We now have the possibility of adding a new directed link e_8 with capacity $c_8 = 1$ and we are given $x > 0$ extra units of capacity to distribute on the links (including e_8). The goal is to decide where to place e_8 and how to allocate the additional

capacity in order to maximize the throughput from o to d, and to study the maximum throughput $T(x)$ as a function of $x \geq 0$.

From Part (b), without the new link, the maximum throughput is limited by the bottleneck formed by the links (a, d) and (c, d) in the minimum cut. Adding the new edge e_8 allows us to partially bypass this bottleneck.

The most effective choice is to place the new link directly between the origin and the destination,

$$e_8 = (o, d).$$

Any o–d cut must separate o from d, so the edge (o, d) crosses every possible cut. Increasing its capacity therefore increases the capacity of all cuts simultaneously. In this configuration the initial minimum cut capacity becomes 6, so $T(0) = 6$, and it is optimal to allocate all the extra capacity x to the new link (o, d). This yields a simple linear relation between the throughput and the available extra capacity:

$$T_{\text{best}}(x) = 6 + x, \qquad x \geq 0.$$

For comparison, if we place the new link internally, for example as $e_8 = (b, d)$, it only appears in some of the cuts. In this case the throughput still increases with x, but more slowly. For $x = 0, 1, \ldots, 12$ we obtain

$$T_{\text{internal}}(x) = 6,\ 6,\ 7,\ 7,\ 8,\ 8,\ 9,\ 9,\ 10,\ 10,\ 11,\ 11,\ 12,$$

which corresponds to a staircase-shaped curve that grows by one unit of throughput approximately every two units of additional capacity.

The comparison clearly shows that the direct edge $e_8 = (o, d)$ dominates any internal placement: for each value of x the throughput with $e_8 = (o, d)$ is strictly larger than or equal to the throughput obtained with an internal edge. Therefore, to maximize the throughput from o to d we should add the new link as (o, d) and allocate all the extra capacity to this link, obtaining

$$T(x) = 6 + x, \qquad x \geq 0.$$

I. Exercise 2

Part (a)

In this part we compute the Katz centrality of all nodes in the graph of Figure 2, using

$$\beta = 0.15,$$

together with a uniform intrinsic centrality vector μ, meaning that every node receives the same baseline importance.

Katz centrality assigns higher values to nodes that are not only well connected but are also linked to other important nodes. Formally, it is defined as the solution of

$$x = (I - \beta A)^{-1}\mu,$$

where A is the adjacency matrix of the graph. The parameter β must be chosen smaller than the inverse of the largest eigenvalue of A, and the value $\beta = 0.15$ satisfies this condition for our graph.

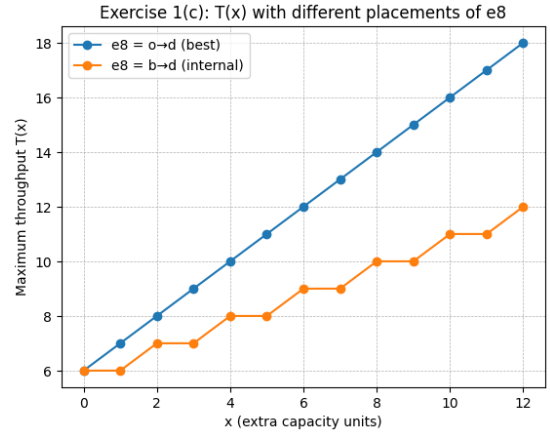

Fig. 2. Maximum throughput $T(x)$ as a function of the extra capacity x for two different placements of the new link e_8: a direct link (o, d) and an internal link (b, d).

After constructing the network exactly as in Figure 2 and computing the Katz scores, we observe a clear pattern that reflects the structure of the graph:

- **Node** $n6$ receives the highest centrality value. This is expected because it acts as a major hub connecting two large parts of the graph: the clique on the right and the branching structure on the left. Being adjacent to several important nodes greatly increases its Katz score.
- **Nodes** $n1$ **to** $n5$ also obtain high and almost identical centrality values. They form a clique, so each of them is connected to many influential neighbors, which naturally results in a high Katz centrality.
- **Node** $n9$ shows a moderate centrality. It is the center of a star and lies on the path connecting the left part of the graph to the rest. Although it has many neighbors, most of them are peripheral nodes with low importance, which explains why its value is lower than that of the clique nodes.
- **Nodes** $n10$–$n14$ have the lowest centrality values since they are leaf nodes connected only to $n9$.

The two nodes highlighted in the assignment have Katz centrality values:

$$n6 = 1.0000, \qquad n9 = 0.4715,$$

confirming that $n6$ is the most influential node in the graph, whereas $n9$ plays an intermediate role between the central and peripheral regions of the network.

Part (b)

In this part we design a distributed algorithm to compute the PageRank centrality of the graph in Figure 2, using a damping parameter $\beta = 0.15$ and a uniform intrinsic centrality vector μ.

PageRank can be defined as the fixed point of

$$x = \beta\mu + (1 - \beta)Px,$$

where P is a column-stochastic matrix whose entry P_{ji} is equal to $1/\deg(i)$ if there is an edge from node i to node j, and zero otherwise. In our setting the graph is undirected and the intrinsic centrality is uniform, so $\mu_i = 1/n$ for all nodes i.

This equation can be solved by an iterative scheme that is naturally distributed. Each node i maintains a local estimate $x_i(k)$ of its PageRank at iteration k, initialized as $x_i(0) = 1/n$. At every iteration, node i gathers the current values of its neighbors $x_j(k)$ and their degrees $\deg(j)$, and updates its own value according to

$$x_i(k+1) = \beta\frac{1}{n} + (1-\beta)\sum_{j\in\mathcal{N}(i)}\frac{x_j(k)}{\deg(j)},$$

where $\mathcal{N}(i)$ denotes the set of neighbors of node i. This local update rule only requires information exchanged along edges: in each round every node sends the quantity $x_i(k)/\deg(i)$ to its neighbors and then updates its own state. Therefore, the algorithm can be implemented in a fully distributed way.

The iterations are repeated until convergence, for example when

$$\|x(k+1) - x(k)\|_1 < \text{tol},$$

for a prescribed tolerance `tol`. The limit vector x represents the PageRank centrality of all nodes.

Applying this distributed PageRank algorithm to our graph with $\beta = 0.15$ and uniform μ yields the highest centrality for node $n9$, followed by node $n6$, then the clique nodes $n1, \ldots, n5$, and finally the leaf nodes in the left-hand star. In particular, we obtain

$$\text{PR}(n9) \approx 0.1940, \qquad \text{PR}(n6) \approx 0.1145,$$

showing that $n9$ concentrates the largest amount of PageRank probability in the network.

Part (c)

The Katz and PageRank centralities highlight two different notions of importance in the graph. Katz centrality assigns the highest value to node $n6$, while node $n9$ receives a significantly smaller score. This is because Katz rewards global reachability: node $n6$ occupies a strategic bridging position, connecting the dense clique ($n1$–$n5$) with the rest of the network and linking also to $n15$. As a result, many short and long walks in the graph pass through or arrive at $n6$, which increases its Katz score. In contrast, node $n9$ is the center of a small star whose leaves are relatively isolated from the main part of the graph, limiting its global accessibility.

PageRank produces the opposite ranking: node $n9$ becomes more central than $n6$. This is due to the way PageRank distributes importance. Each of the five degree-one nodes $n10$–$n14$ sends its entire PageRank mass directly to $n9$, giving $n9$ a strong and undiluted inflow. Node $n6$, instead, receives contributions mainly from clique nodes that split their importance among many neighbors, resulting in diluted inflows. Consequently, PageRank highlights the local hub structure of $n9$, whereas Katz emphasizes the global bridging role of $n6$.

Part (d)

We compute the PageRank centrality of nodes $n6$ and $n9$ for

$$\beta \in \{0,\, 0.25,\, 0.5,\, 0.75,\, 1\},$$

using a uniform intrinsic vector μ. The resulting values are

β	$PR(n6)$	$PR(n9)$	$\Delta(\beta) = PR(n6) - PR(n9)$
0	0.1458	0.1250	+0.0208
0.25	0.1106	0.1911	−0.0805
0.5	0.1014	0.1667	−0.0653
0.75	0.0875	0.1270	−0.0395
1.0	0.0667	0.0667	0

These values show that $\Delta(\beta)$ is not monotone: it is positive at $\beta = 0$, negative for intermediate β, and returns to zero at $\beta = 1$.

For $\beta = 0$, PageRank coincides with the stationary distribution of a simple random walk, which in undirected graphs is proportional to node degree. Since $\deg(n6) = 7$ and $\deg(n9) = 6$, we obtain $PR(n6) > PR(n9)$.

For $\beta = 1$, the process becomes pure teleportation and the PageRank vector is uniform, so $PR(n6) = PR(n9)$.

For intermediate values $0 < \beta < 1$, node $n9$ becomes more central because its five degree-one neighbors $n10$–$n14$ pass their entire PageRank mass directly to $n9$, producing a strong and undiluted inflow. Node $n6$, instead, receives diluted inflows from high-degree clique nodes that distribute their importance among many neighbors. This makes $PR(n9) > PR(n6)$ for a range of intermediate β.

In conclusion, $n6$ dominates when link structure drives the dynamics, $n9$ dominates for intermediate β due to concentrated inflows, and both nodes become equal when $\beta = 1$.

II. Exercise 3

Part (a)

In this part we determine the shortest path from Santa Monica (node 1) to Santa Ana (node 17). Since the network is assumed to be empty, the travel time on each link is equal to its minimum value l_e, so finding the fastest route is equivalent to solving a shortest–path problem with edge weights given by l_e.

Using the link travel times provided in `traveltime.mat`, we run a shortest–path algorithm on the directed network. The algorithm selects the route that minimizes the sum of the minimum travel times along the path. The unique path with minimum total cost is

$$1 \to 2 \to 3 \to 9 \to 13 \to 17.$$

This path corresponds to travelling from Santa Monica through Los Angeles and Norwalk and then directly to Santa Ana. Figure 3 highlights this route on the network: all other links are shown in light grey, while the links $(1,2)$, $(2,3)$, $(3,9)$, $(9,13)$ and $(13,17)$ are emphasized as the fastest path.

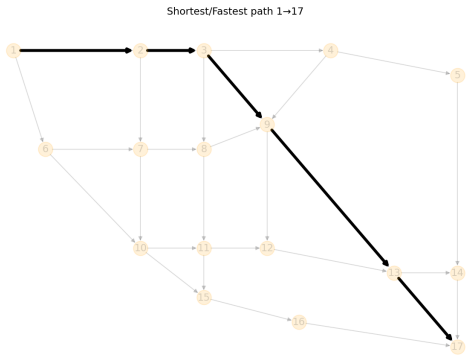

Fig. 3. Shortest/fastest path from node 1 to node 17 in the empty network.

Part (b)

In this part we compute the maximum flow that can be sent from Santa Monica (node 1) to Santa Ana (node 17). The directed network is the same as in Part (a), but now each link e_i is endowed with a capacity c_{e_i} taken from the vector `capacities.mat`. We treat the capacities as upper bounds on the amount of traffic that can be carried by each highway segment.

We model the problem as a standard single-source single-sink maximum flow problem with source node 1, sink node 17 and arc capacities c_e. Using a max–flow algorithm (NetworkX's `maximum_flow` implementation), we obtain the value of the maximum feasible flow from node 1 to node 17 as

$$\text{MaxFlow}(1 \to 17) = 22448.$$

By the Max–Flow Min–Cut theorem, this value is equal to the capacity of the minimum 1–17 cut in the network.

Part (c)

We are given a feasible flow vector $f \in \mathbb{R}^{28}$ (stored in `flow.mat`) and we first compute the corresponding node imbalance vector ν by

$$\nu = Bf,$$

where B is the node–link incidence matrix. The i-th component ν_i represents the net external inflow at node i: a positive value means that node i must receive flow from outside the network in order to sustain the link flows, whereas a negative value means that node i sends flow to the outside.

Multiplying the incidence matrix by the given flow vector yields

$$\nu = Bf = \begin{bmatrix} 16806 \\ 8570 \\ 19448 \\ 4957 \\ -746 \\ 4768 \\ 413 \\ -2 \\ -5671 \\ 1169 \\ -5 \\ -7131 \\ -380 \\ -7412 \\ -7810 \\ -3430 \\ -23544 \end{bmatrix}.$$

As expected, the sum of all components of ν is zero, which reflects conservation of flow at the network level. In the remainder of the exercise, we simplify the setting by assuming that the exogenous inflow is zero at all nodes except node 1 and node 17. We keep the total injected flow equal to the inflow at node 1 computed above, and define

$$\nu_1 = 16806, \qquad \nu_{17} = -16806, \qquad \nu_i = 0 \ \text{ for all } i \neq 1, 17.$$

Thus, node 1 acts as the unique source and node 17 as the unique sink, while flow is conserved at all intermediate nodes.

Part (d)

We compute the socially optimal flow f by minimizing the total travel time in the network. Each link e has delay

$$\tau_e(f_e) = \frac{l_e}{1 - f_e/c_e},$$

so the total system cost is

$$J(f) = \sum_{e \in E} f_e \tau_e(f_e) = \sum_{e \in E} \left(\frac{l_e c_e}{1 - f_e/c_e} - l_e c_e \right).$$

The social optimum is obtained by solving the convex program

$$\begin{aligned} \min_f \quad & J(f) \\ \text{s.t.} \quad & Bf = \nu, \\ & 0 \le f_e \le c_e, \quad e \in E, \end{aligned}$$

where B is the incidence matrix and ν is the inflow vector with node 1 as source and node 17 as sink.

Using `cvxpy`, we obtain an optimal flow vector f with total cost

$$J(f) \approx 2.61 \times 10^4.$$

The optimal flow satisfies all capacity and conservation constraints. Full numerical values of f may be reported in an appendix if required.

Part (e)

We compute the Wardrop equilibrium flow $f^{(0)}$ using the Beckmann formulation. Wardrop's principle states that, at equilibrium, no driver can reduce their travel time by changing routes. The equilibrium can be found by minimizing the potential function

$$\sum_{e\in E}\int_0^{f_e}\tau_e(s)\,ds,$$

subject to flow conservation and capacity constraints. With

$$\tau_e(s)=\frac{l_e}{1-s/c_e},$$

the integral becomes

$$\int_0^{f_e}\tau_e(s)\,ds=-\,l_e c_e\log\left(1-\frac{f_e}{c_e}\right),$$

which is convex for $0\le f_e<c_e$.

Thus the Wardrop equilibrium is obtained by solving

$$\begin{aligned}\min_f\quad & \sum_{e\in E}-\,l_e c_e\log\left(1-\frac{f_e}{c_e}\right)\\ \text{s.t.}\quad & Bf=\nu,\\ & 0\le f_e\le c_e,\quad e\in E,\end{aligned}$$

where B is the incidence matrix and ν is the inflow vector with node 1 as source and node 17 as sink.

Using `cvxpy`, we obtain a Wardrop equilibrium flow $f^{(0)}$ with total potential value

$$\Phi\big(f^{(0)}\big)\approx 1.57\times 10^4.$$

The equilibrium flow satisfies all conservation and capacity constraints. Full numerical values of $f^{(0)}$ may be reported in an appendix if needed.

Part (f)

We now introduce link tolls in order to align individual selfish behavior with the social optimum. For each link e, the toll is defined as

$$\omega_e=\psi_e'(f_e^*)-\tau_e(f_e^*),$$

where $\psi_e(f_e)=f_e\tau_e(f_e)$ is the contribution of link e to the social cost and f_e^* is the flow at the system optimum found in Part (d). For the considered $\psi_e(\cdot)$ this simplifies to

$$\omega_e=f_e^*\,\tau_e'(f_e^*),$$

so each toll is equal to the marginal external delay imposed on the system by an additional unit of flow on that link.

With these tolls, the perceived delay on link e becomes $\tau_e(f_e)+\omega_e$. The corresponding Wardrop equilibrium $f(\omega)$ can again be computed using the Beckmann formulation, now with objective

$$\sum_{e\in E}\left(\int_0^{f_e}\tau_e(s)\,ds+\omega_e f_e\right),$$

subject to the same flow conservation and capacity constraints as before. We solve this convex program using `cvxpy`.

Numerically, the resulting equilibrium flow $f(\omega)$ is almost identical to the social optimum f^*: the maximum absolute difference between the components of $f(\omega)$ and f^* is of the order of 1, while typical flows are of order 10^4. Hence, up to numerical precision,

$$f(\omega)\approx f^*.$$

This confirms the theoretical result that, with appropriately chosen marginal-cost tolls, the Wardrop equilibrium of the tolled network coincides with the system optimal flow.

Part (g)

We now change the system cost from the total travel time to the total *additional* travel time with respect to free flow. For each link e we set

$$\psi_e(f_e)=f_e\big(\tau_e(f_e)-l_e\big),$$

so that the system cost is $\sum_{e\in E}\psi_e(f_e)$. Using the delay function $\tau_e(f_e)=l_e/(1-f_e/c_e)$, this yields a convex objective in f, and the social optimum is obtained by solving

$$\begin{aligned}\min_f\quad & \sum_{e\in E}\psi_e(f_e)\\ \text{s.t.}\quad & Bf=\nu,\\ & 0\le f_e\le c_e,\quad e\in E.\end{aligned}$$

Solving this convex program in `cvxpy`, we obtain a new system-optimal flow f^* and a total additional cost

$$\sum_{e\in E}\psi_e(f_e^*)\approx 1.53\times 10^4.$$

Following the instructions, we then construct a toll vector ω^* such that

$$\omega_e^*=\psi_e'(f_e^*)-\tau_e(f_e^*).$$

For the chosen $\psi_e(\cdot)$ this simplifies to

$$\omega_e^*=-\,l_e+f_e^*\,\tau_e'(f_e^*).$$

With these tolls, the perceived delay on link e becomes $\tau_e(f_e)+\omega_e^*$.

We compute the Wardrop equilibrium $f(\omega^*)$ of the tolled network using again the Beckmann formulation, that is, by minimizing

$$\sum_{e\in E}\Big(\int_0^{f_e}\tau_e(s)\,ds+\omega_e^* f_e\Big)$$

subject to the same flow conservation and capacity constraints. Numerically, the resulting equilibrium flow $f(\omega^*)$ is extremely close to the new system optimum f^*: the maximum absolute difference between the components is of order 1, while typical flows are of order 10^3–10^4. Hence, up to numerical precision,

$$f(\omega^*)\approx f^*.$$

This shows that, also under the alternative system cost based on additional travel time, appropriately chosen tolls can induce a Wardrop equilibrium that coincides with the system-optimal flow.